

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T36 · Graphing Inequalities

With Answers

Questions with full worked solutions

■ 31 pages

FAMILY MAP

Graphing Inequalities

F01

Recover inequalities from a shaded region

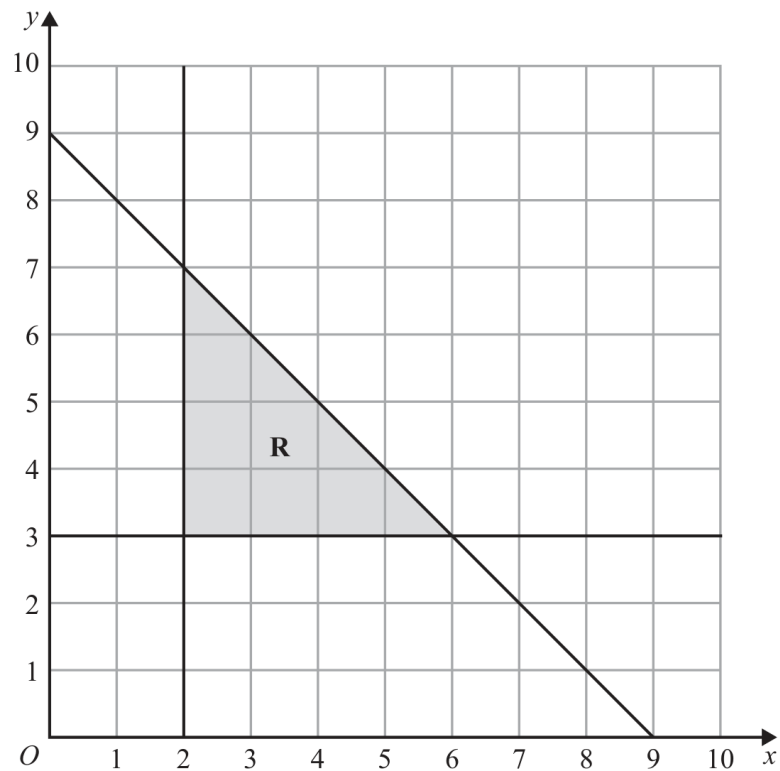
5 Q | 16 marks

F02

Construct the feasible region from inequalities

8 Q | 26 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

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(3)

Worked Solution 25

Region of inequalities

UNIT 2**3 marks****Read the three boundaries**

$$x = 2, \quad y = 3, \quad x + y = 9.$$

Choose the shaded side

$$R : \quad x \geq 2, \quad y \geq 3, \quad x + y \leq 9.$$

Final answer

$$x \geq 2, \quad y \geq 3, \quad x + y \leq 9$$

Question 13

4MA1-2H Q2

ORIGINAL QUESTION

4 marks

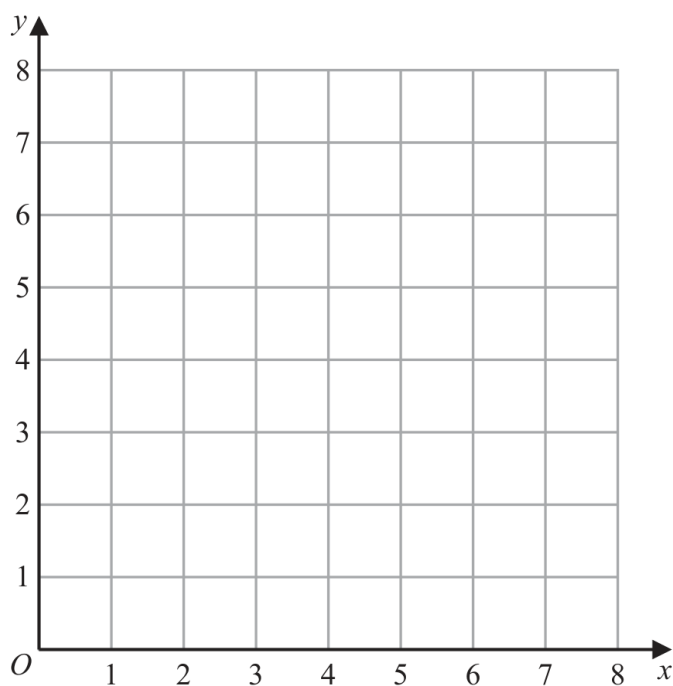
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$y \geq 2$$

$$x \leq 6$$

$$y \leq x + 1$$

Label the region **R**

(1)

Worked Solution 13

Lines and a region of inequalities

MODULAR UNIT 2

4 marks

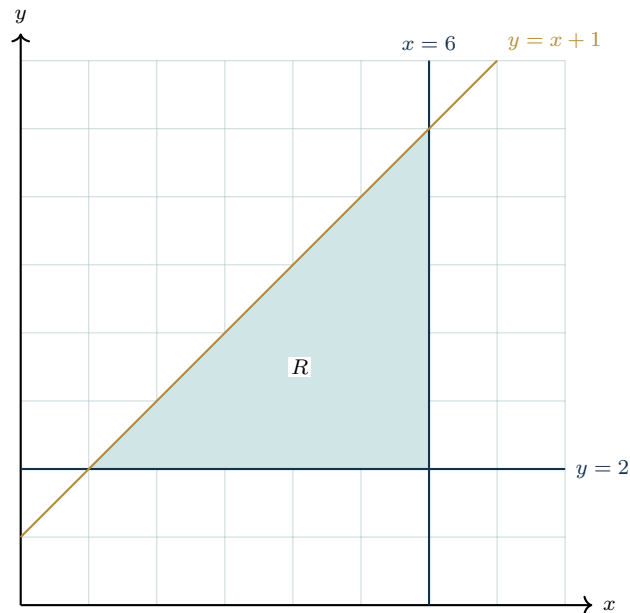
BoundariesDraw and label $y = 2$, $x = 6$, $y = x + 1$.**Region**Shade the closed triangle with vertices $(1, 2)$, $(6, 2)$, $(6, 7)$ and label R .**Final answer**The triangular region R

Completed diagram 13

Lines and a region of inequalities

MODULAR UNIT 2

4 marks

Completed solution diagram: Lines and feasible region

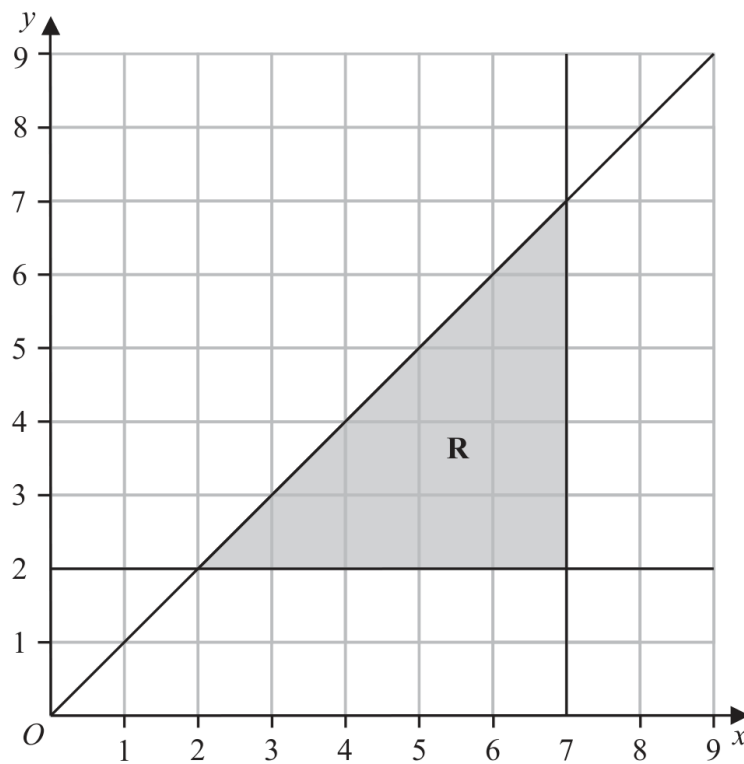
Question 05

4MA1-1H Q8 (b)

ORIGINAL QUESTION UNIT 2

3 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

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(3)

Worked solution 05

Define the shaded region

MODULAR UNIT 2

3 marks

(b) – Define the shaded region

Read each boundary

$$y = 2, \quad x = 7, \quad y = x.$$

Choose the shaded side

$$R : y \geq 2, x \leq 7, y \leq x.$$

Final answer

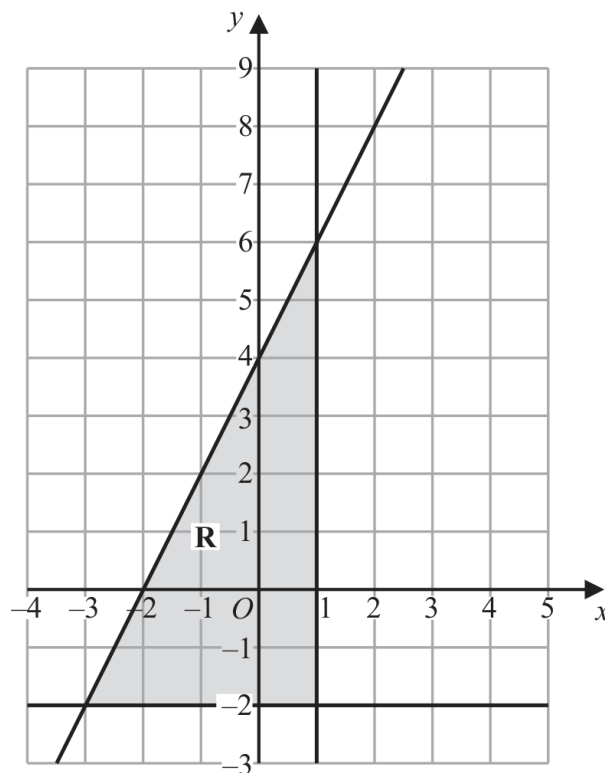
$$y \geq 2, x \leq 7, y \leq x$$

Question 06

4MA1-1H Q7

ORIGINAL QUESTION

4 marks



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

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Worked Solution 06

Region inequalities

MODULAR UNIT 2

4 marks

Read the vertical boundary

$$x \leq 1.$$

Read the horizontal boundary

$$y \geq -2.$$

Find the sloping line and shaded side

$$y = 2x + 4, \quad y \leq 2x + 4.$$

Final answer

$$x \leq 1, \quad y \geq -2, \quad y \leq 2x + 4$$

Original question 21

4MA1-2H Q7 M01, M02

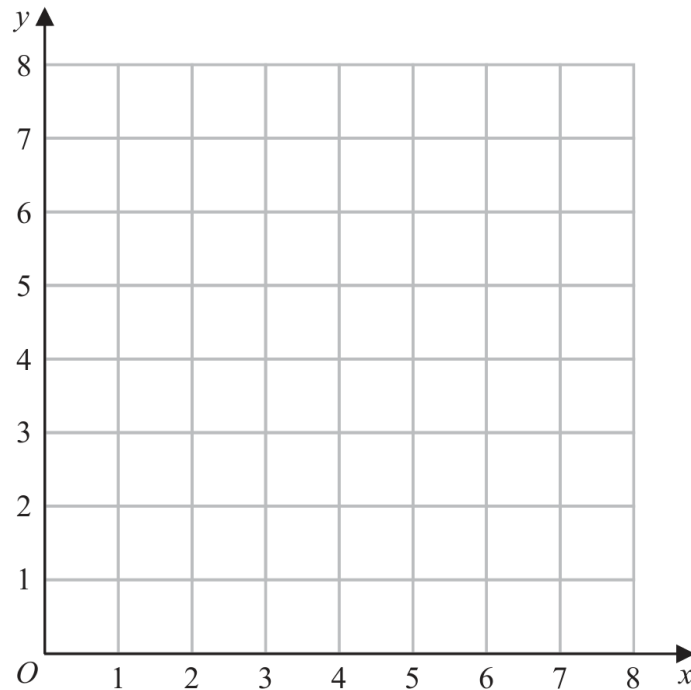
MODULAR UNIT 2

4 marks

7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geq 3 \quad y \geq 1 \quad x + y \leq 7$$

Label the region **R**

(1)

Worked solution 21

Draw three straight lines / Shade the simultaneous-inequality region

MODULAR UNIT 2

4 marks

M01 – Draw three straight lines**Draw $x=3$** A vertical line through $x=3$.**Draw $y=1$** A horizontal line through $y=1$.**Draw $x+y=7$**

A line of gradient -1 through the intercepts (0,7) and (7,0); label all three.

Final answerThe labelled lines $x=3$, $y=1$ and $x+y=7$ **M02 – Shade the simultaneous-inequality region****Translate each inequality**

$$x \geq 3, \quad y \geq 1, \quad x + y \leq 7.$$

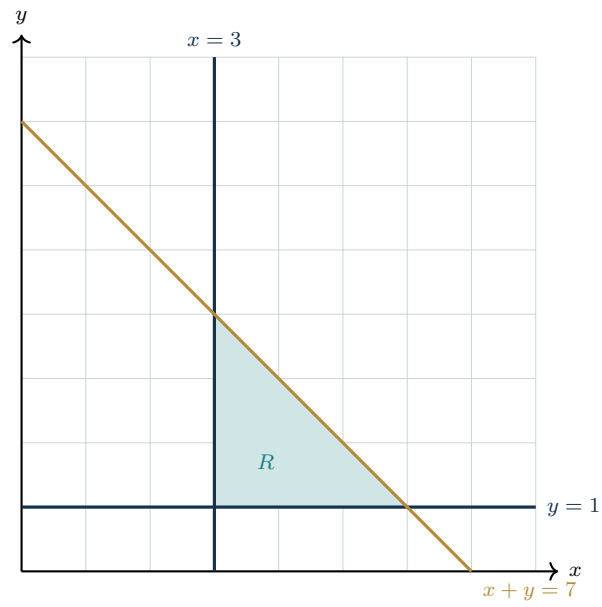
Shade and label

Shade the closed triangular region and label it R.

Final answer

$$R : x \geq 3, y \geq 1, x + y \leq 7$$

Completed solution diagram: Boundary lines and the Unit 2 feasible region



Original question 21

4MA1-2H Q2 Q2(a), Q2(b)

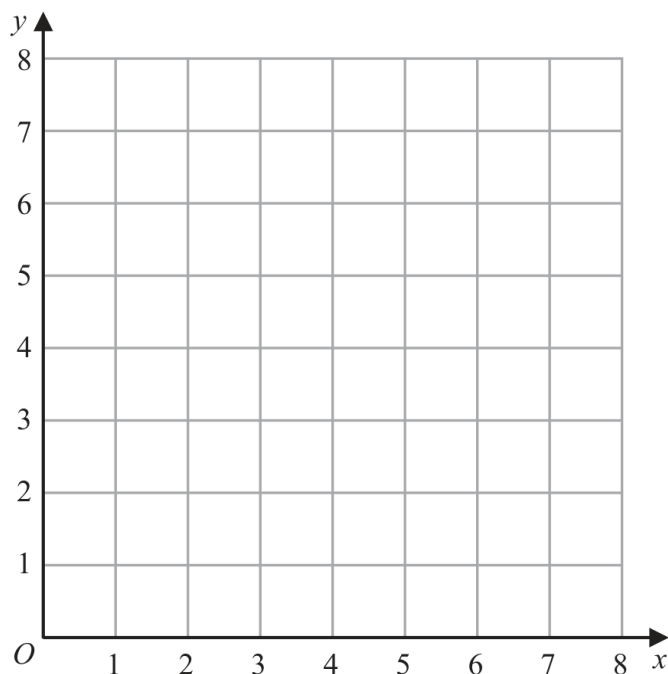
ORIGINAL QUESTION**4 marks**

- 2 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$

(ii) $x = 2$

(iii) $x + y = 7$



(3)

- (b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$

$x \geq 2$

$x + y \leq 7$

Label the region **R**.

(1)

Worked solution 21Chapter: Linear Graphs ($y = mx + c$); Graphing Inequalities

MODULAR UNIT 2

4 marks

Q2(a) – Chapter: Linear Graphs ($y = mx + c$)

Family: Generate and plot a linear graph • Variant: Plot a line from generated values

MethodDraw and label the horizontal line $y = 1$.*Why: Every point on this line has y -coordinate 1.***Method**Draw and label the vertical line $x = 2$.*Why: Every point on this line has x -coordinate 2.***Method**Draw and label $x + y = 7$ through $(0, 7)$ and $(7, 0)$.*Why: Two intercepts determine the straight line.***Final answer**line 1: $y = 1$ **Final answer**line 2: $x = 2$ **Final answer**line 3: $x + y = 7$ **Q2(b) – Chapter: Graphing Inequalities**

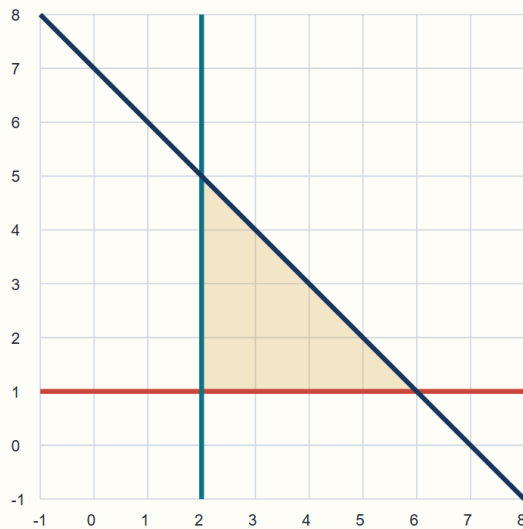
Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

MethodShade the region above $y = 1$, to the right of $x = 2$, and below $x + y = 7$; label it R.*Why: All three inequalities must hold simultaneously.***Final answer****region:** R has vertices $(2, 1)$, $(2, 5)$, $(6, 1)$

Completed diagram 21Chapter: Linear Graphs ($y = mx + c$); Graphing Inequalities

MODULAR UNIT 2

4 marks

Q2 Completed Cartesian graphBoundary lines for $y=1$, $x=2$ and $x+y=7$ 

$$y = 1$$

$$x = 2$$

$$x + y = 7$$

Region: $x \geq 2$, $y \geq 1$, $x+y \leq 7$

Vertices: (2,1), (6,1), (2,5)

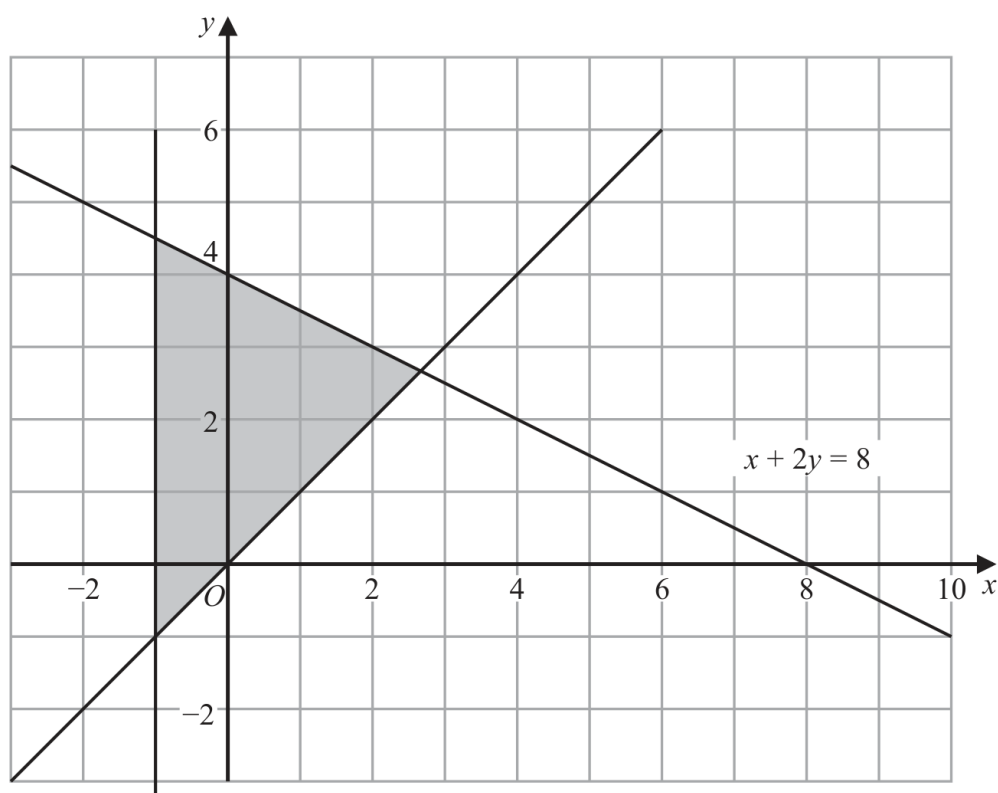
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Original question 27

4MA1-2H Q13

ORIGINAL QUESTION**3 marks**

- 13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

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Worked solution 27

Chapter: Graphing Inequalities

MODULAR UNIT 2

3 marks

Q13 – Chapter: Graphing Inequalities

Family: Recover inequalities from a shaded region • Variant: Derive the sloping boundary equation from the graph

Method

The vertical boundary is $x = -1$ and the shaded side is $x \geq -1$.

Why: Test a point in the shaded region.

Method

The rising boundary is $y = x$ and the shaded side is $y \geq x$.

Why: Read the line and shaded side.

Method

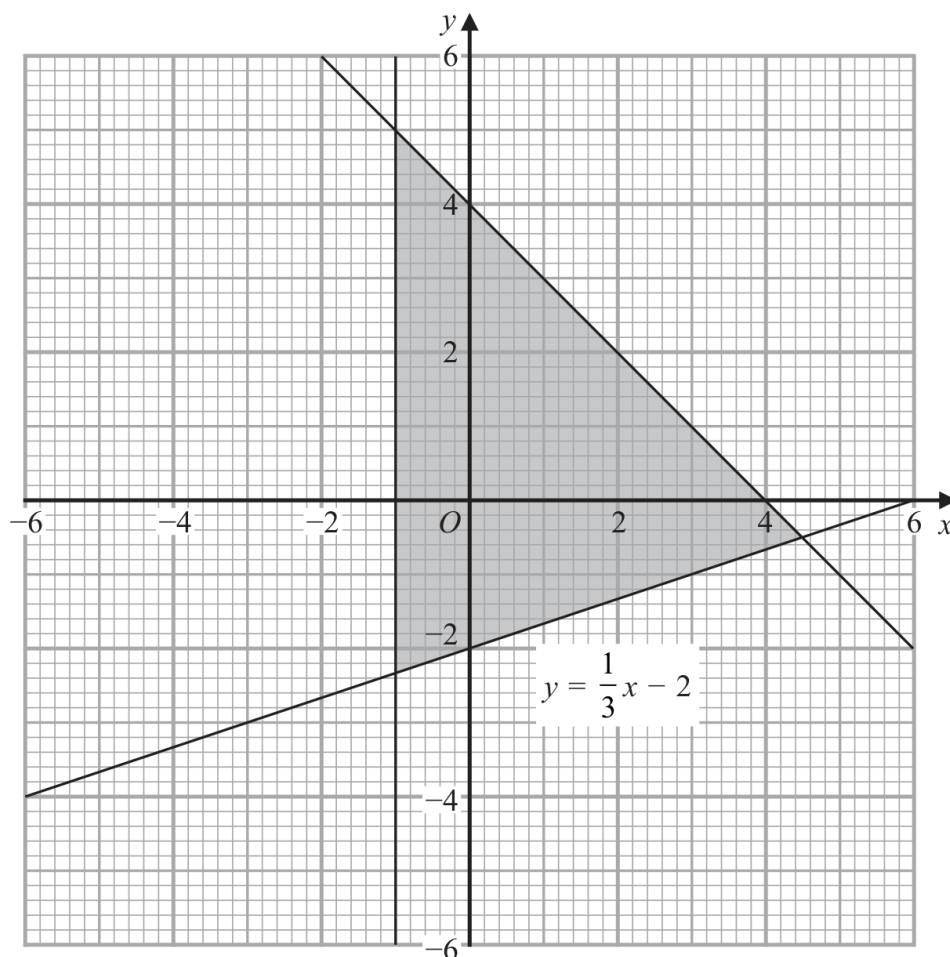
The falling boundary is $x + 2y = 8$ and the shaded side is $x + 2y \leq 8$.

Why: Use the labelled boundary equation and a test point.

Final answer

inequalities: $x \geq -1$, $y \geq x$, $x + 2y \leq 8$

- 10 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

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Worked solution

January 2020 | Question 10 | 3 marks

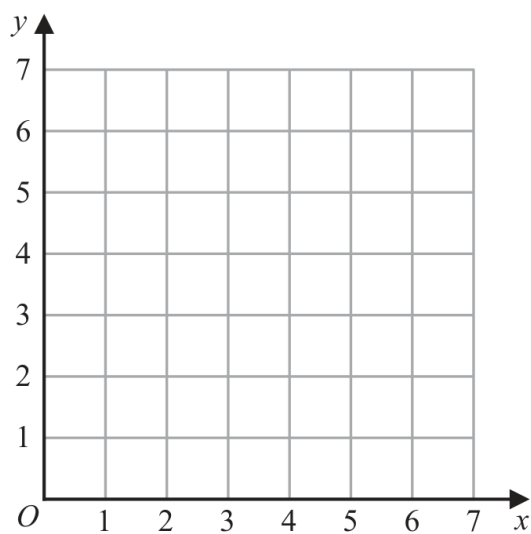
Family: Recover inequalities from a shaded region • Variant: Derive the sloping boundary equation from the graph

Method / working

1. The vertical boundary is $x=-1$ and the shaded side is to the right: $x \geq -1$.
2. The oblique boundary $x+y=4$ has the shaded side below it: $x+y \leq 4$.
3. The given boundary $y=(1/3)x-2$ has the shaded side above it: $y \geq (1/3)x-2$.

Final answer: $x \geq -1$; $x+y \leq 4$; $y \geq (1/3)x-2$

4



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

Worked solution

January 2021 | Question 4 | 3 marks

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Method / working

1. Draw the vertical line $x=1.5$.
2. Draw $y=x$ through the origin.
3. Draw $x+y=6$ through $(0,6)$ and $(6,0)$, labelling each line.

Final answer: Draw $x=1.5$, $y=x$ and $x+y=6$

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$$x \geq 1.5$$

$$y \geq x$$

$$x + y \leq 6$$

Label the region **R**.

(1)

Worked solution

January 2021 | Question 4 | 1 marks

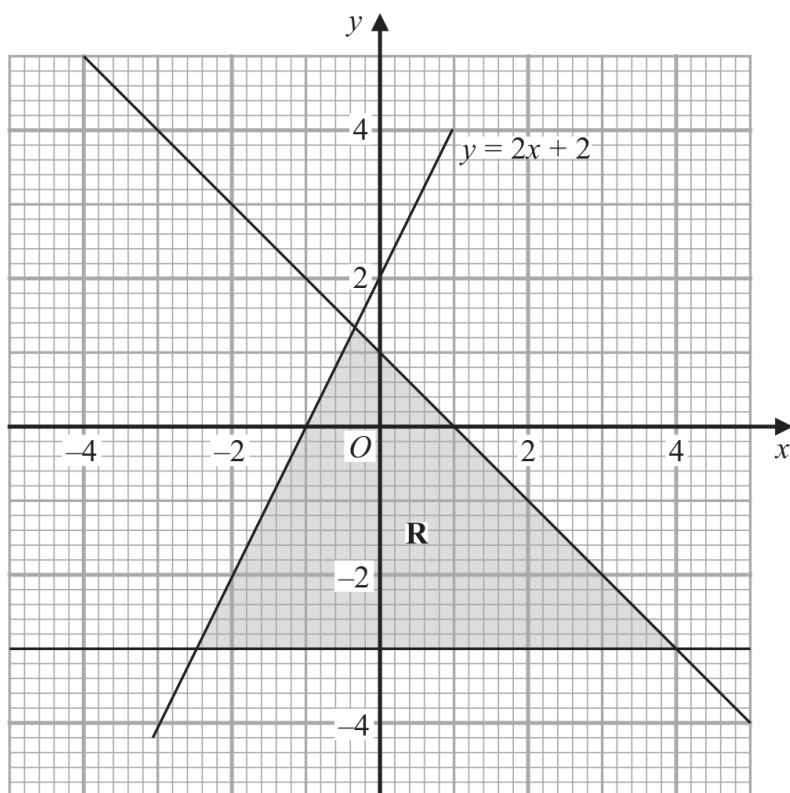
Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Method / working

1. Shade the intersection of the three half-planes and label it R.

Final answer: Region R: $x \geq 1.5$, $y \geq x$, $x + y \leq 6$

13



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Write down the three inequalities that define **R**.

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Worked solution

November 2021 | Question 13 | 3 marks

Family: Recover inequalities from a shaded region | Variant: Derive the sloping boundary equation from the graph

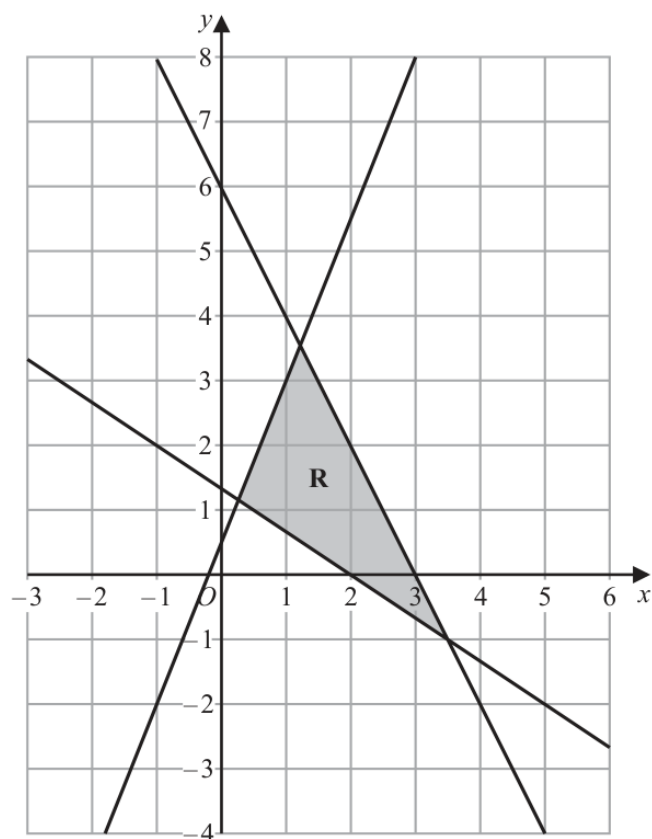
METHOD / WORKING

1. The bottom boundary is
2. $y = -3$
3. so
4. $y \geq -3$
5. The rising boundary is
6. $y = 2x + 2$
7. so the region has
8. $y \leq 2x + 2$
9. The falling boundary goes through $(0, 1)$ and $(1, 0)$, so
10. $x + y = 1$
11. The region is below this line:
12. $x + y \leq 1$
13. Answer: $y \geq -3, y \leq 2x + 2, x + y \leq 1$.

FINAL ANSWER

 $y \geq -3, y \leq 2x + 2, x + y \leq 1$.

11



The region **R**, shown shaded in the diagram, is bounded by the straight lines with equations

$$2x + y = 6$$

$$2y = 5x + 1$$

$$3y + 2x = 4$$

Write down the three inequalities that define **R**

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(Total for Question 11 is 3 marks)

Answer reference | January 2023 | Question 11

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

11		$2x + y \leq 6$ $2y \leq 5x + 1$ $3y + 2x \geq 4$	3	B3 oe for all three correct (B2 oe for any two correct) (B1 oe for any one correct) $2x + y \leq 6$ equivalent to $y \leq -2x + 6$ oe $2y \leq 5x + 1$ equivalent to $y \leq 2.5x + 0.5$ oe $3y + 2x \geq 4$ equivalent to $y \geq -\frac{2}{3}x + \frac{4}{3}$ oe Allow the following inequalities $2x + y < 6$ oe $2y < 5x + 1$ oe $3y + 2x > 4$ oe
				Total 3 marks

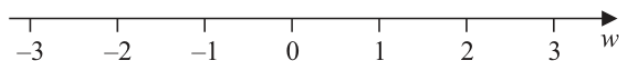
Worked solution

January 2023 | Question 11 | 3 marks

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

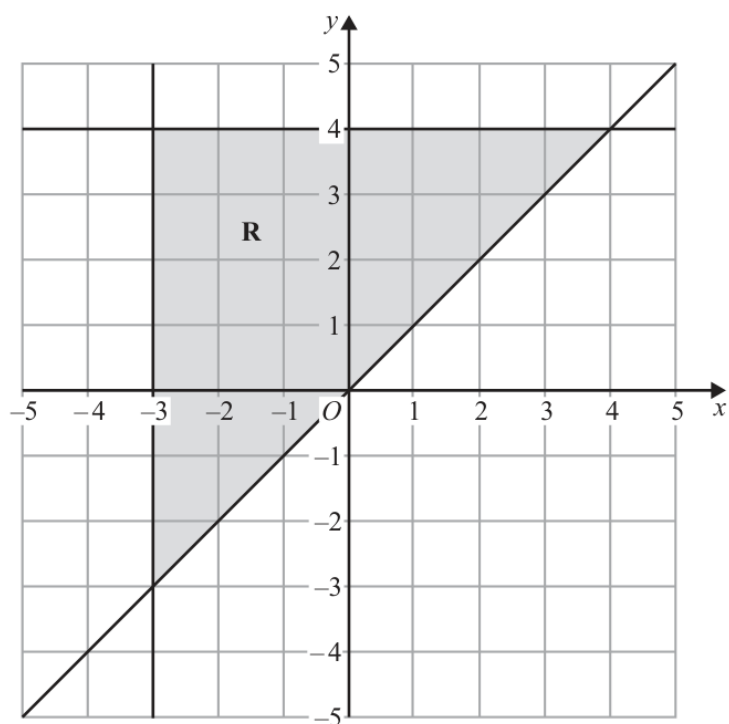
Official final mark-scheme pages are embedded immediately after this question.

- 10 (a) On the number line, represent the inequality $w < 1$



(1)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

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(3)

(Total for Question 10 is 4 marks)

Worked solution

November 2025 | Question 10 | 4 marks

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Solution

For part (a), $w < 1$ is shown with an open circle at 1 and an arrow to the left.

For part (b), the region is right of $x = -3$, below $y = 4$, and above $y = x$:

$$x \geq -3$$

$$y \leq 4$$

$$y \geq x$$

Answer: (a) Open circle at 1, shade left. (b) $x \geq -3$, $y \leq 4$, $y \geq x$.